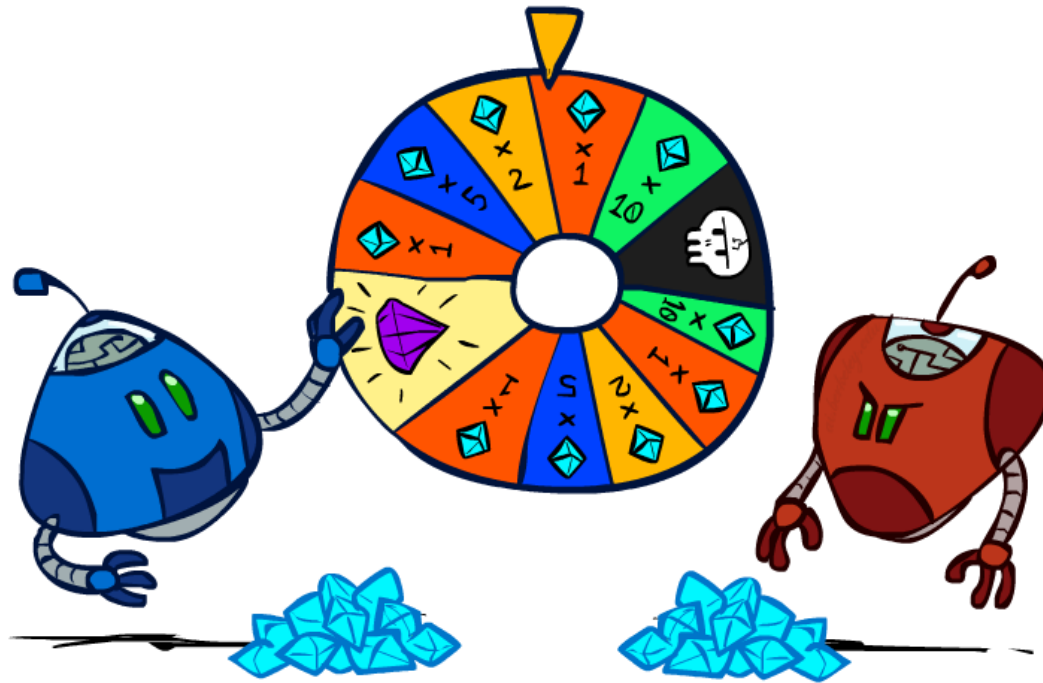


CS 188: Artificial Intelligence

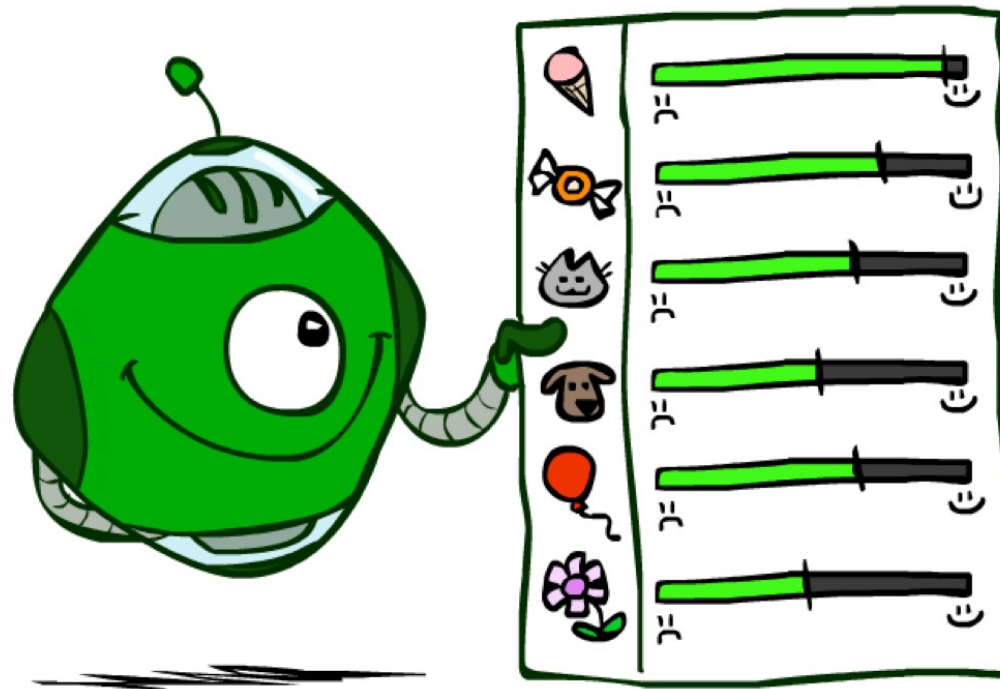
Rational Decisions



Slides from Stuart Russell and Peyrin Kao

University of California, Berkeley

Utilities



Utilities

- Utilities are functions from outcomes (states of the world) to real numbers that describe an agent's preferences
- Where do utilities come from?
 - In a game, may be simple (+1/-1)
 - Utilities summarize the agent's goals
 - Theorem: any "rational" preferences can be summarized as a utility function
- We hard-wire utilities and let behaviors emerge
 - Why don't we hard-wire behaviors?

because there are too much states and our memory cannot hold it

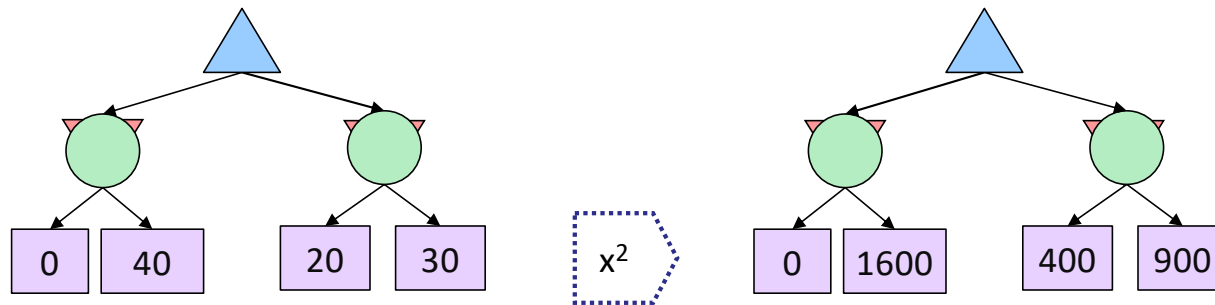


Maximum Expected Utility

- Principle of maximum expected utility:
 - A rational agent should choose the action that **maximizes its expected utility, given its knowledge**
- Questions:
 - Where do utilities come from?
 - How do we know such utilities even exist?
 - How do we know that averaging makes sense?
 - What if our behavior (preferences) can't be described by utilities?



Utility magnitudes are meaningful

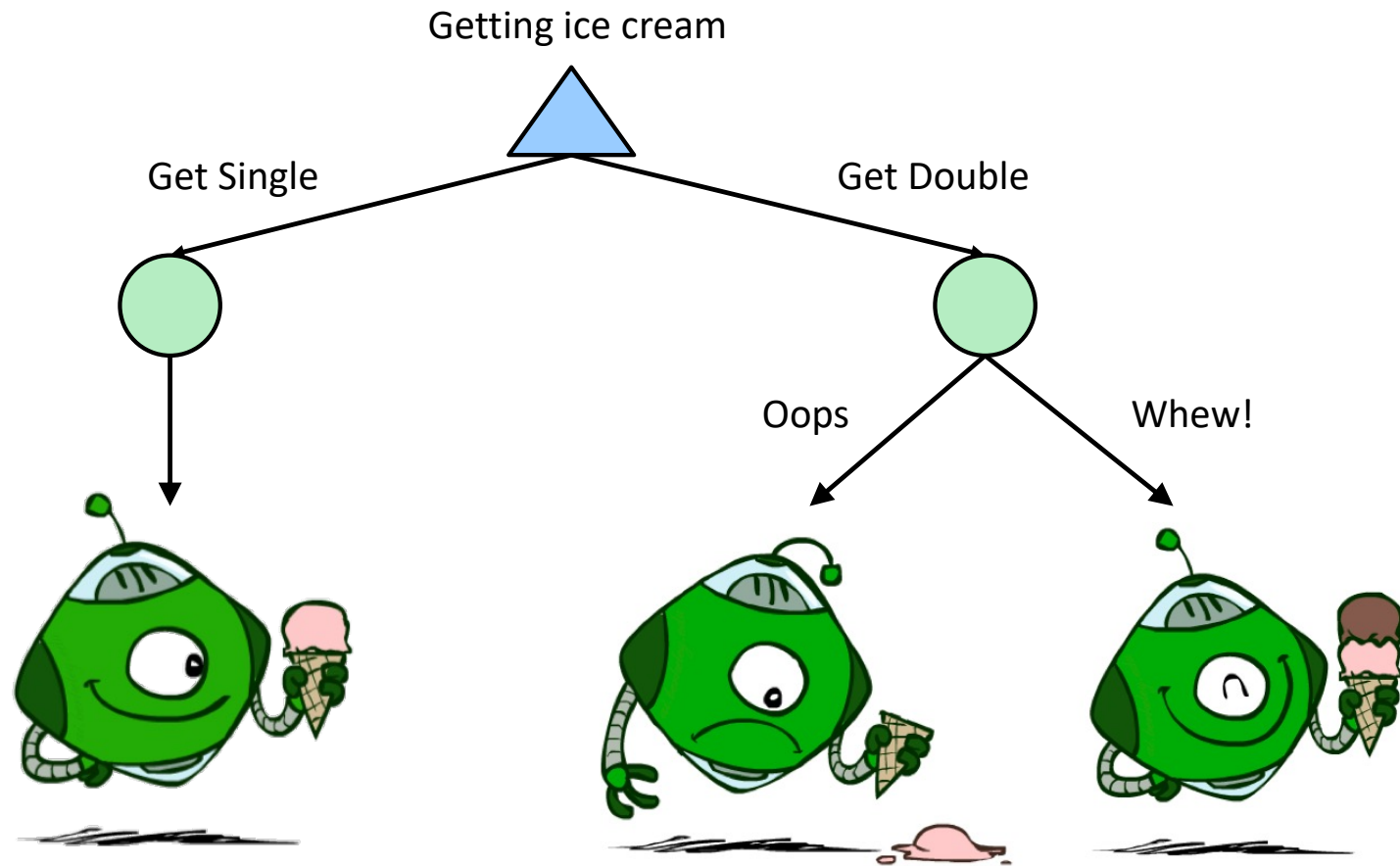


- For worst-case minimax reasoning, terminal value scale doesn't matter
 - We just want better states to have higher evaluations (get the ordering right)
 - The optimal decision is invariant under any **monotonic transformation**
- For average-case expectimax reasoning, we need **magnitudes** to be meaningful

较优选择是不变的，在任何单调映射的情况下

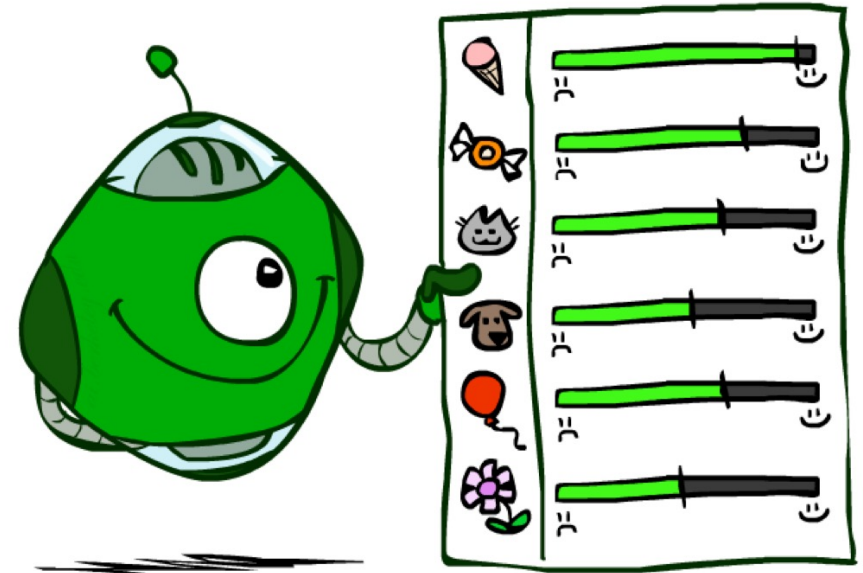
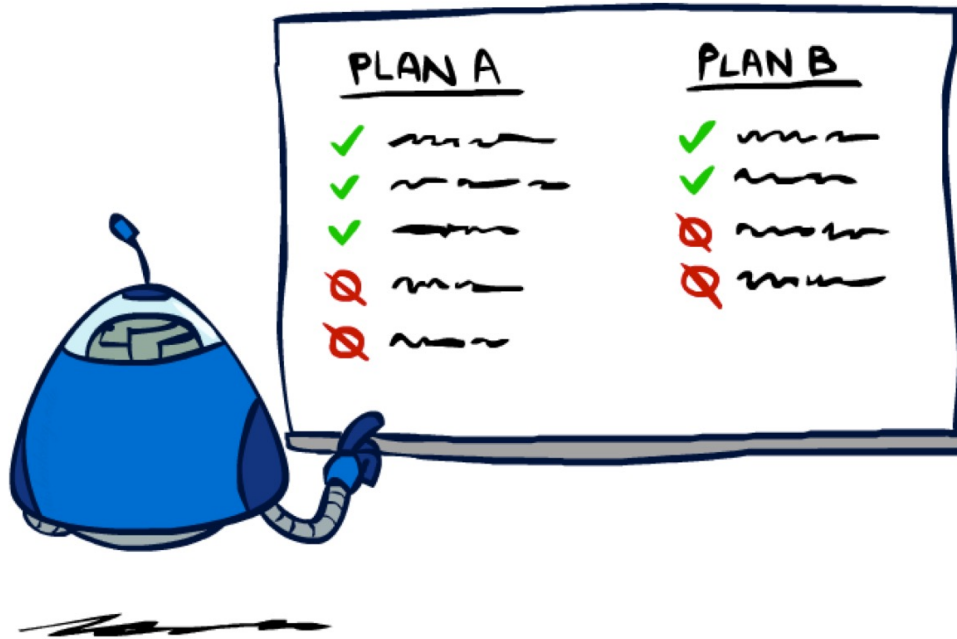
average case means we get reasoning by synthesis all the outcomes
which means the magnitudes can have an impact on last decision, instead of just ranking

Utilities: Uncertain Outcomes



我们希望最终得到一个“效用”，也就是从理性偏好下得到一个合理的评分函数

Deriving Utilities from Rational Preferences



Preferences

- An agent must have preferences among:

- Prizes: A , B , etc.
- Lotteries: situations with uncertain prizes

$$L = [p, A; (1-p), B]$$

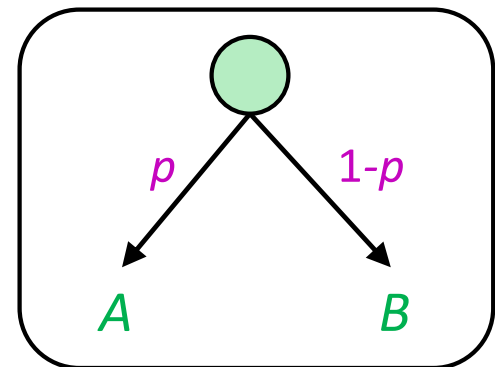
- Notation:

- Preference: $A > B$
- Indifference: $A \sim B$

A Prize



A Lottery

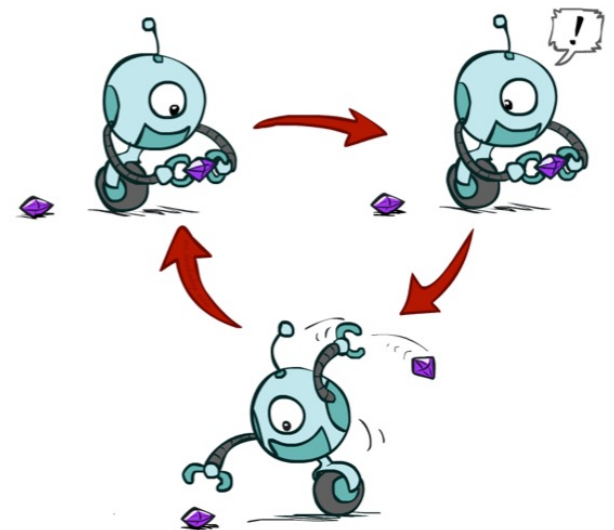


Rational Preferences

- We want some constraints on preferences before we call them rational, such as:

Axiom of Transitivity: $(A > B) \wedge (B > C) \Rightarrow (A > C)$

- Costs of irrationality:
- An agent with **intransitive preferences** can be induced to give away all of its money
 - If $B > C$, then an agent with C would pay (say) 1 cent to get B
 - If $A > B$, then an agent with B would pay (say) 1 cent to get A
 - If $C > A$, then an agent with A would pay (say) 1 cent to get C



Rational Preferences

The Axioms of Rationality

Orderability:

$$(A > B) \vee (B > A) \vee (A \sim B)$$

Transitivity:

$$(A > B) \wedge (B > C) \Rightarrow (A > C)$$

Continuity:

$$(A > B > C) \Rightarrow \exists p [p, A; 1-p, C] \sim B$$

Substitutability:

$$(A \sim B) \Rightarrow [p, A; 1-p, C] \sim [p, B; 1-p, C]$$

Monotonicity:

$$(A > B) \Rightarrow \\ (p \geq q) \Leftrightarrow [p, A; 1-p, B] \geq [q, A; 1-q, B]$$



Theorem: Rational preferences imply behavior describable as maximization of expected utility

MEU Principle

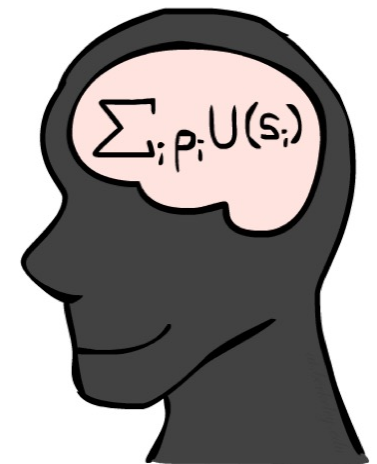
- Theorem [Ramsey, 1931; von Neumann & Morgenstern, 1944]
 - Given any preferences satisfying these constraints, there exists a real-valued function U such that:

$$U(A) \geq U(B) \Leftrightarrow A \geq B$$

类似于数学期望

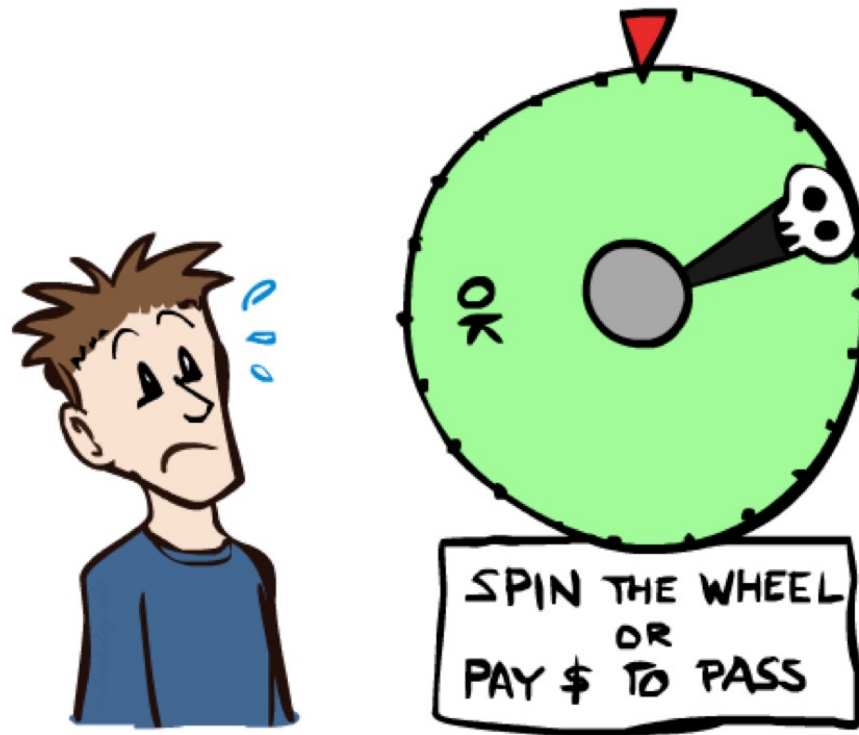
$$U([p_1, S_1; \dots; p_n, S_n]) = p_1 U(S_1) + \dots + p_n U(S_n)$$

- I.e. values assigned by U preserve preferences of both prizes and lotteries!
 - Optimal policy invariant under **positive affine transformation** $U' = aU + b, a > 0$
- Maximum expected utility (MEU) principle:
 - Choose the action that maximizes expected utility 总是选择最大化效用的行为
 - Note: rationality does **not** require representing or manipulating utilities and probabilities
 - E.g., a lookup table for perfect tic-tac-toe



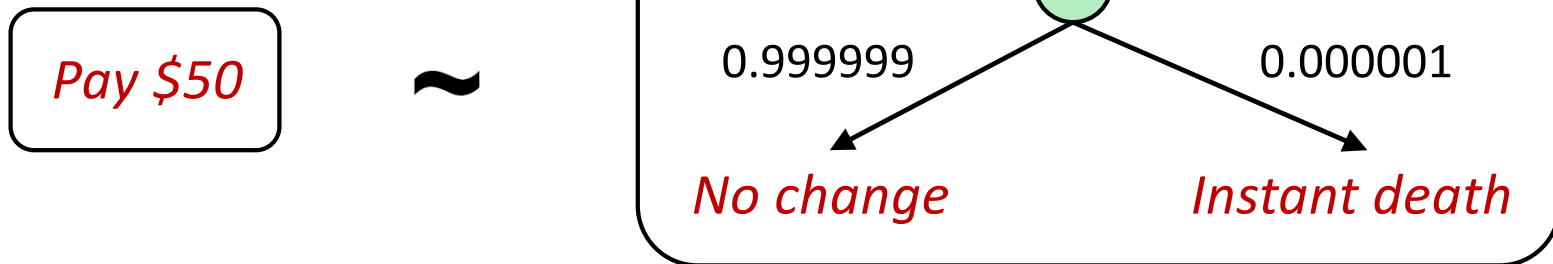
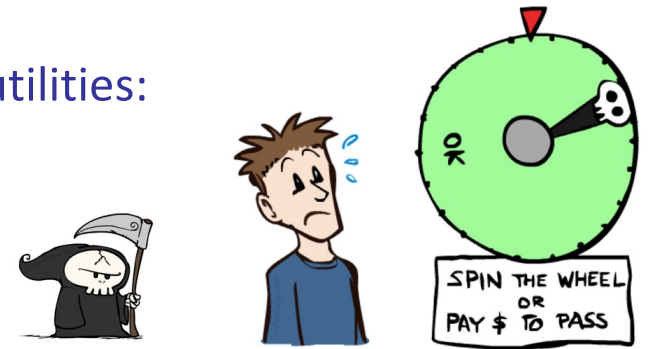
理性行为不等于显式计算：一个表现理性的 Agent，其行为看起来像是正在最大化期望效用，但它不一定在代码里真的写了 U 函数或做了求和运算。例子：一个完美的井字棋（Tic-tac-toe）查找表。它只是机械地查表走棋，没有任何“效用”或“概率”的概念，但因为它的行为永远是最优的，我们可以说它的行为符合 MEU 原则。

Human Utilities



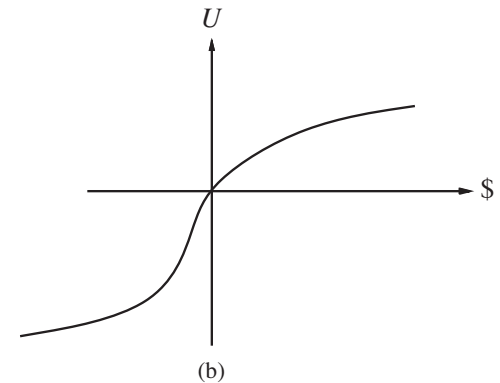
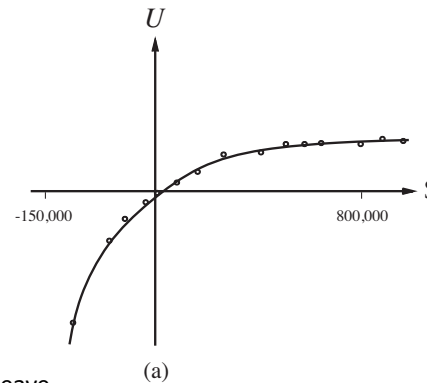
Human Utilities

- Utilities map states to real numbers. Which numbers?
- Standard approach to assessment (elicitation) of human utilities:
 - We want to assign a utility to prize A
 - Compare a prize A to a **standard lottery** L_p between
 - “best possible prize” S_T with probability p
 - “worst possible catastrophe” S_L with probability $1-p$
 - Adjust lottery probability p until indifference: $A \sim L_p$
 - Resulting p is a utility in $[0,1]$

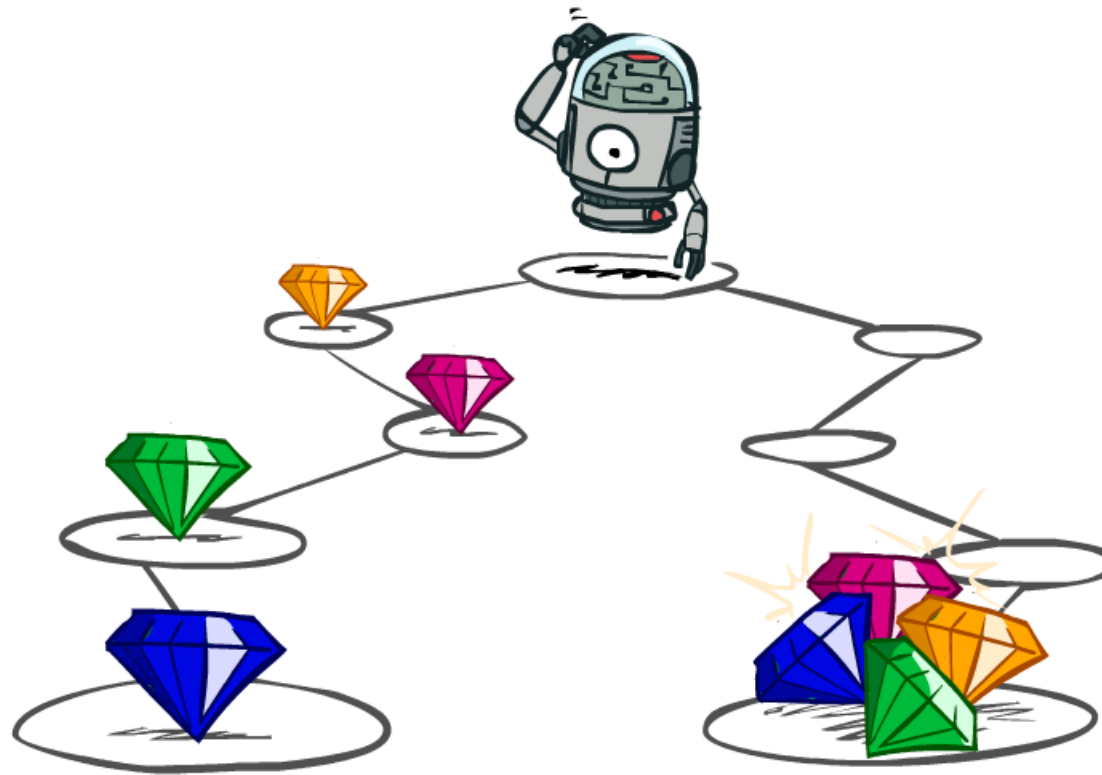


Money

- Money **does not** behave as a utility function, but we can talk about the utility of having money (or being in debt)
- Given a lottery $L = [p, \$X; (1-p), \$Y]$
 - The **expected monetary value** $EMV(L) = pX + (1-p)Y$
 - The utility is $U(L) = pU(\$X) + (1-p)U(\$Y)$
 - Typically, $U(L) < U(EMV(L))$
 - In this sense, people are **risk-averse**
 - E.g., how much would you pay for a lottery ticket $L = [0.5, \$10,000; 0.5, \$0]$?
 - The **certainty equivalent** of a lottery $CE(L)$ is the cash amount such that $CE(L) \sim L$
 - The **insurance premium** is $EMV(L) - CE(L)$
 - If people were risk-neutral, this would be zero!
 - Pay an insurance premium to get out of a lottery
 - House burns down, cybercriminals take your company's data, you die and leave your family with no income

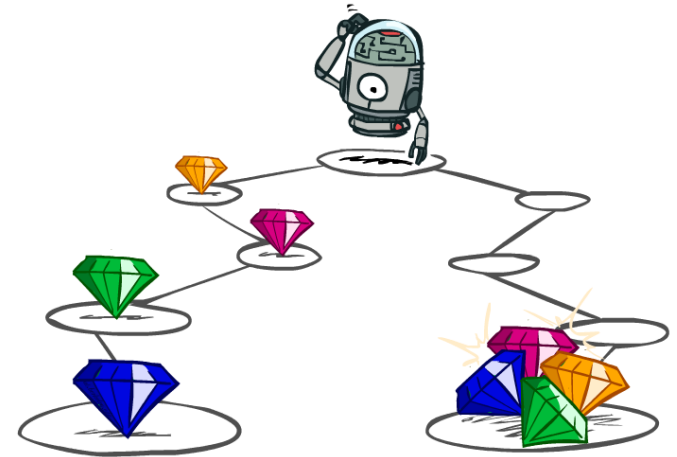


Utilities of Sequences



Utilities of Sequences

- What preferences should an agent have over prize sequences?
- More or less? $[1, 2, 2]$ or $[2, 3, 4]$
- Now or later? $[0, 0, 1]$ or $[1, 0, 0]$



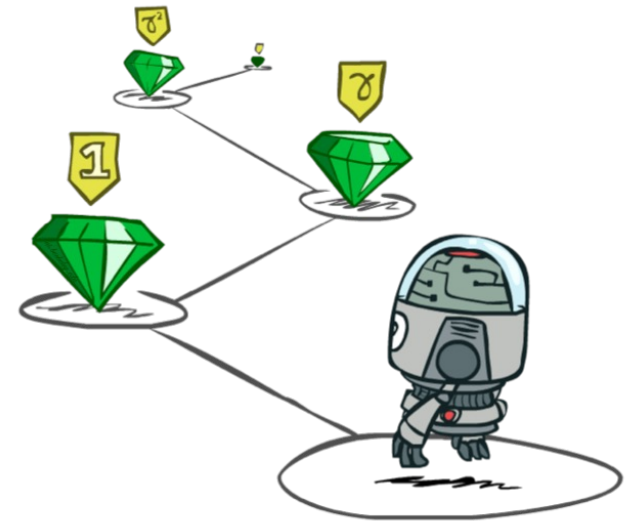
Stationary Preferences

- Theorem: if we assume **stationary preferences**:
 $[a_1, a_2, \dots] > [b_1, b_2, \dots] \Leftrightarrow [c, a_1, a_2, \dots] > [c, b_1, b_2, \dots]$
then there is only one way to define utilities:

- **Additive discounted utility:**

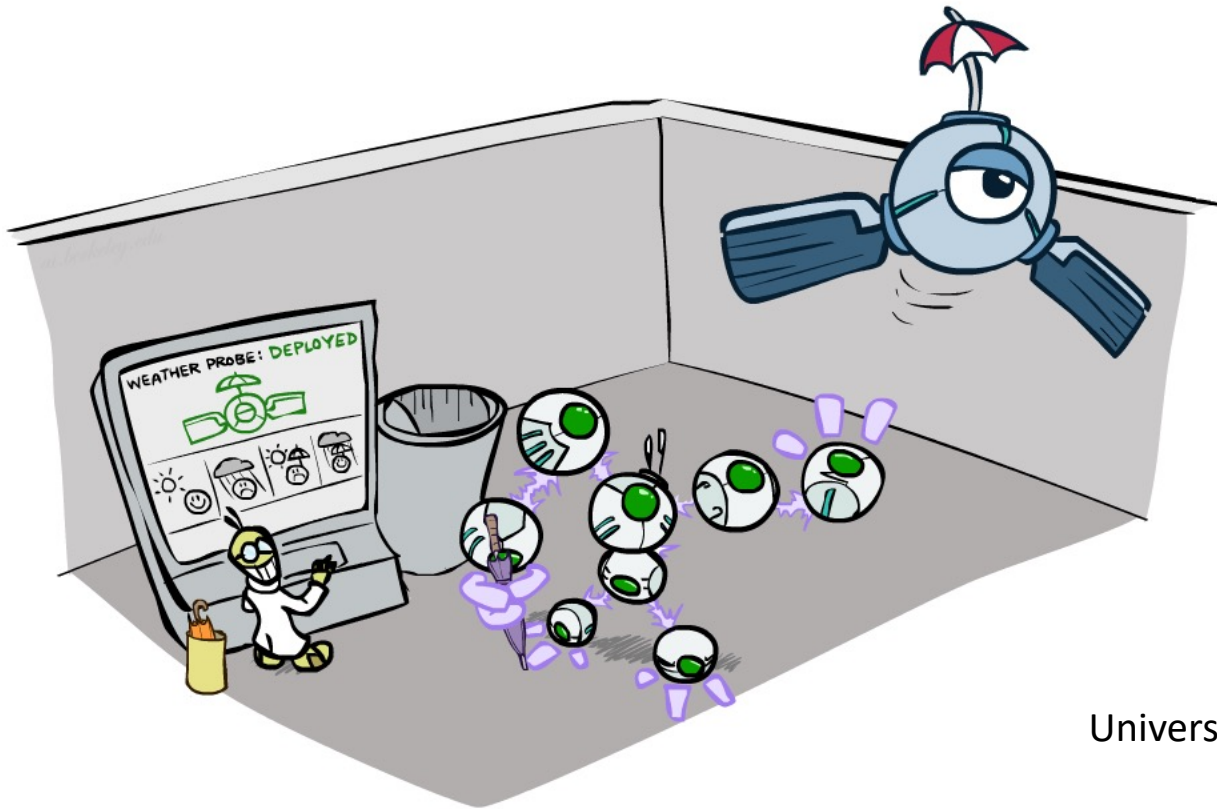
$$U([r_0, r_1, r_2, \dots]) = r_0 + \gamma r_1 + \gamma^2 r_2 + \dots$$

where $\gamma \in (0, 1]$ is the **discount factor**



CS 188: Artificial Intelligence

Decision Networks and Value of Information

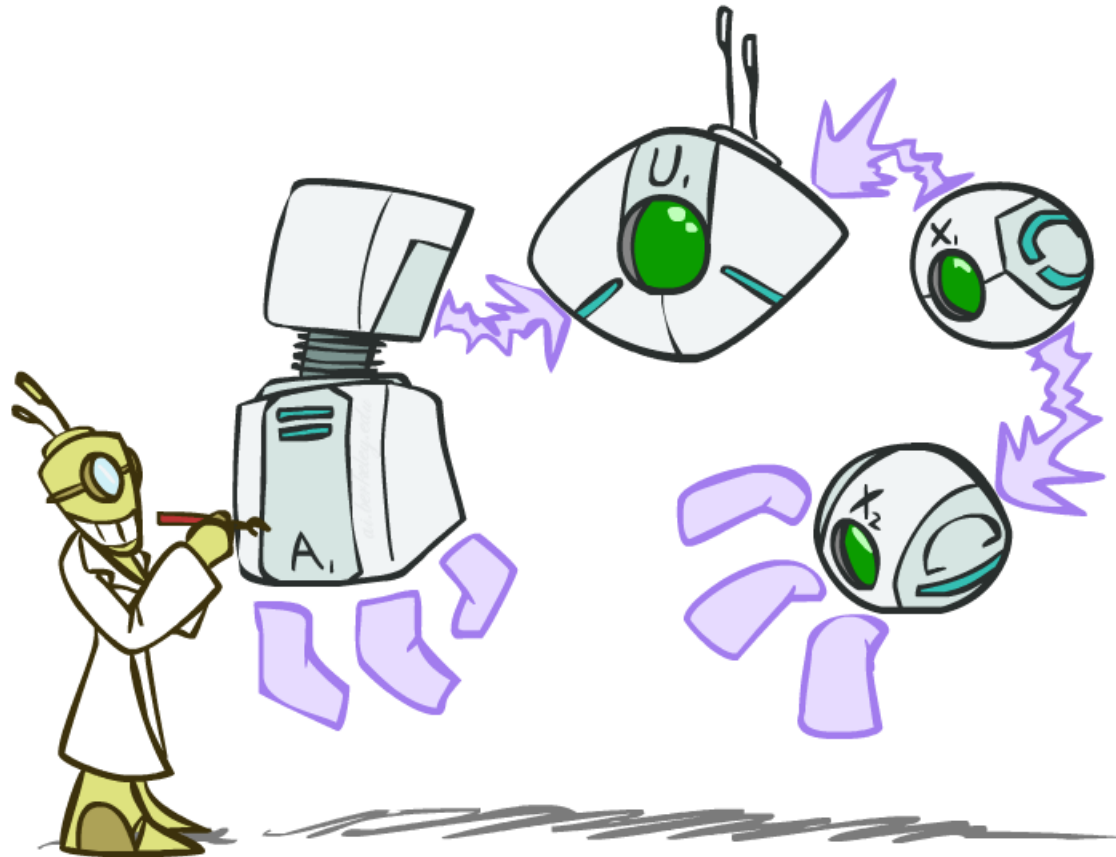


Spring 2024

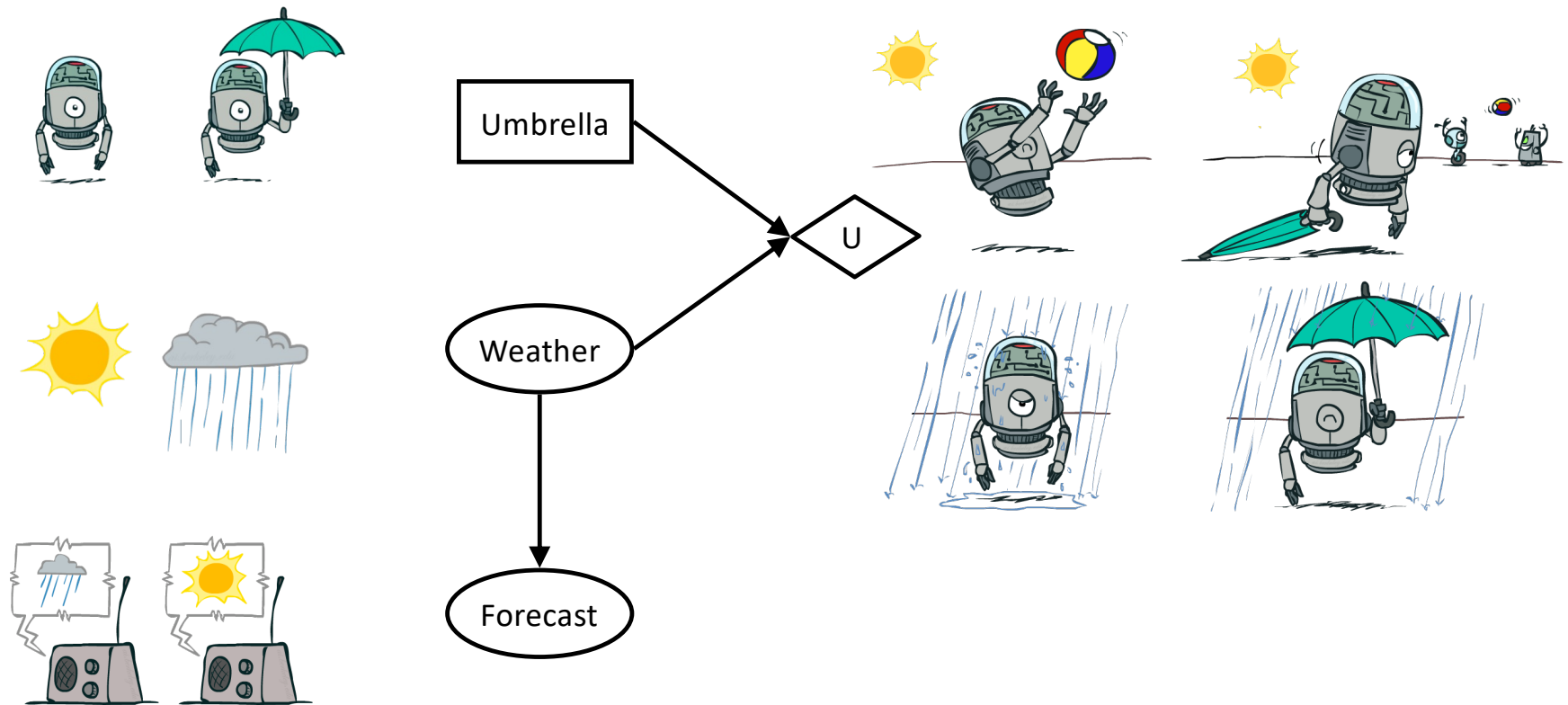
University of California, Berkeley

[These slides were created by Dan Klein and Pieter Abbeel for CS188 Intro to AI at UC Berkeley. All CS188 materials are available at <http://ai.berkeley.edu>.]

Decision Networks



Decision Networks



Decision Networks

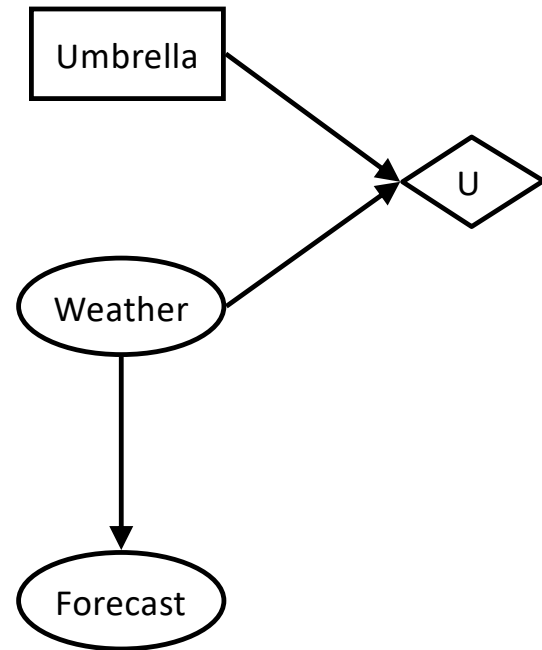
- **MEU: choose the action which maximizes the expected utility given the evidence**

- Can directly operationalize this with decision networks

- Bayes nets with nodes for utility and actions
- Lets us calculate the expected utility for each action

- New node types:

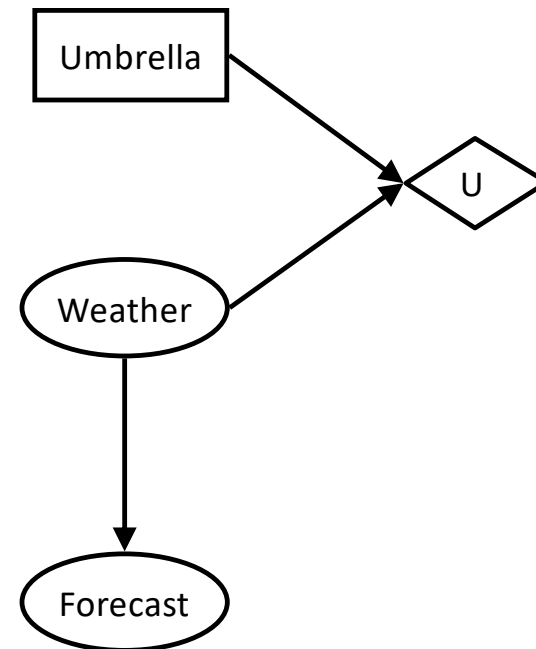
- Chance nodes (just like BNs)
- Actions (rectangles, cannot have parents, act as observed evidence)
- Utility node (diamond, depends on action and chance nodes)



Decision Networks

- Action selection

- Instantiate all evidence
- Set action node(s) each possible way
- Calculate posterior for all parents of utility node, given the evidence
- Calculate expected utility for each action
- Choose maximizing action



Decision Networks

Umbrella = leave

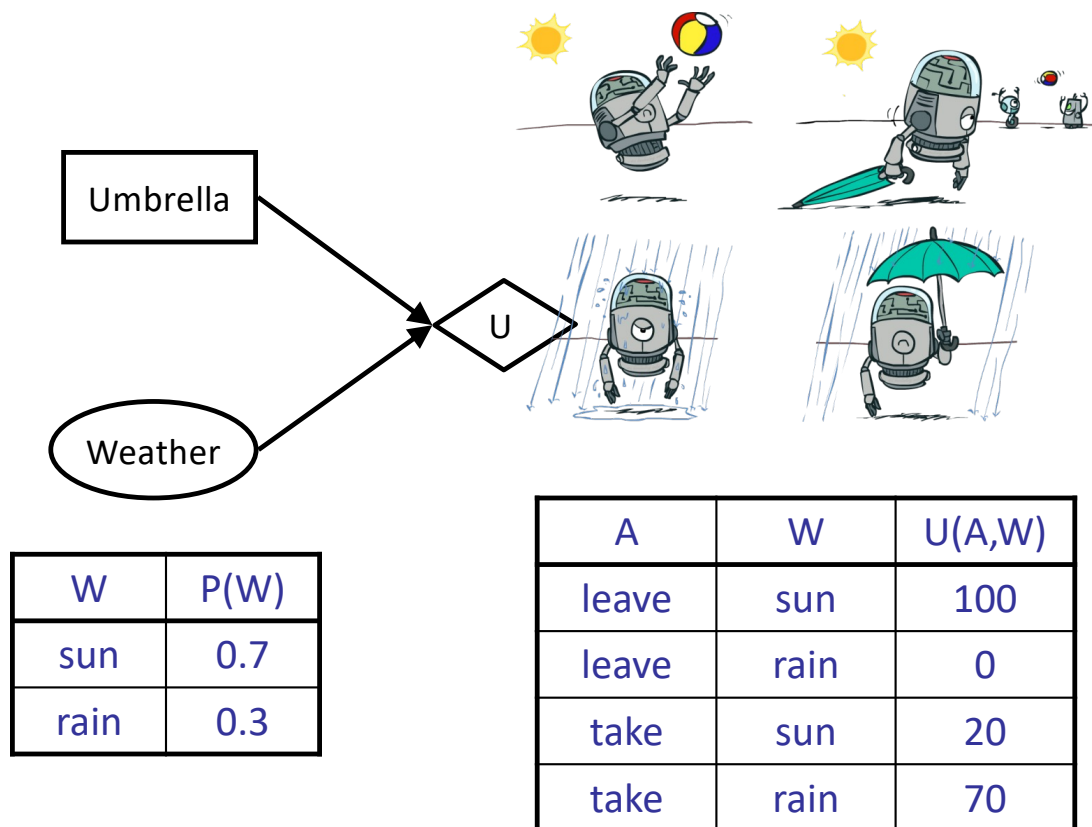
$$\begin{aligned} EU(\text{leave}) &= \sum_w P(w)U(\text{leave}, w) \\ &= 0.7 \cdot 100 + 0.3 \cdot 0 = 70 \end{aligned}$$

Umbrella = take

$$\begin{aligned} EU(\text{take}) &= \sum_w P(w)U(\text{take}, w) \\ &= 0.7 \cdot 20 + 0.3 \cdot 70 = 35 \end{aligned}$$

Optimal decision = leave

$$MEU(\emptyset) = \max_a EU(a) = 70$$



Decision Networks: Notation

Umbrella = leave

$$\begin{aligned} \text{EU}(\text{leave}) &= \sum_w P(w)U(\text{leave}, w) \\ &= 0.7 \cdot 100 + 0.3 \cdot 0 = 70 \end{aligned}$$

Umbrella = take

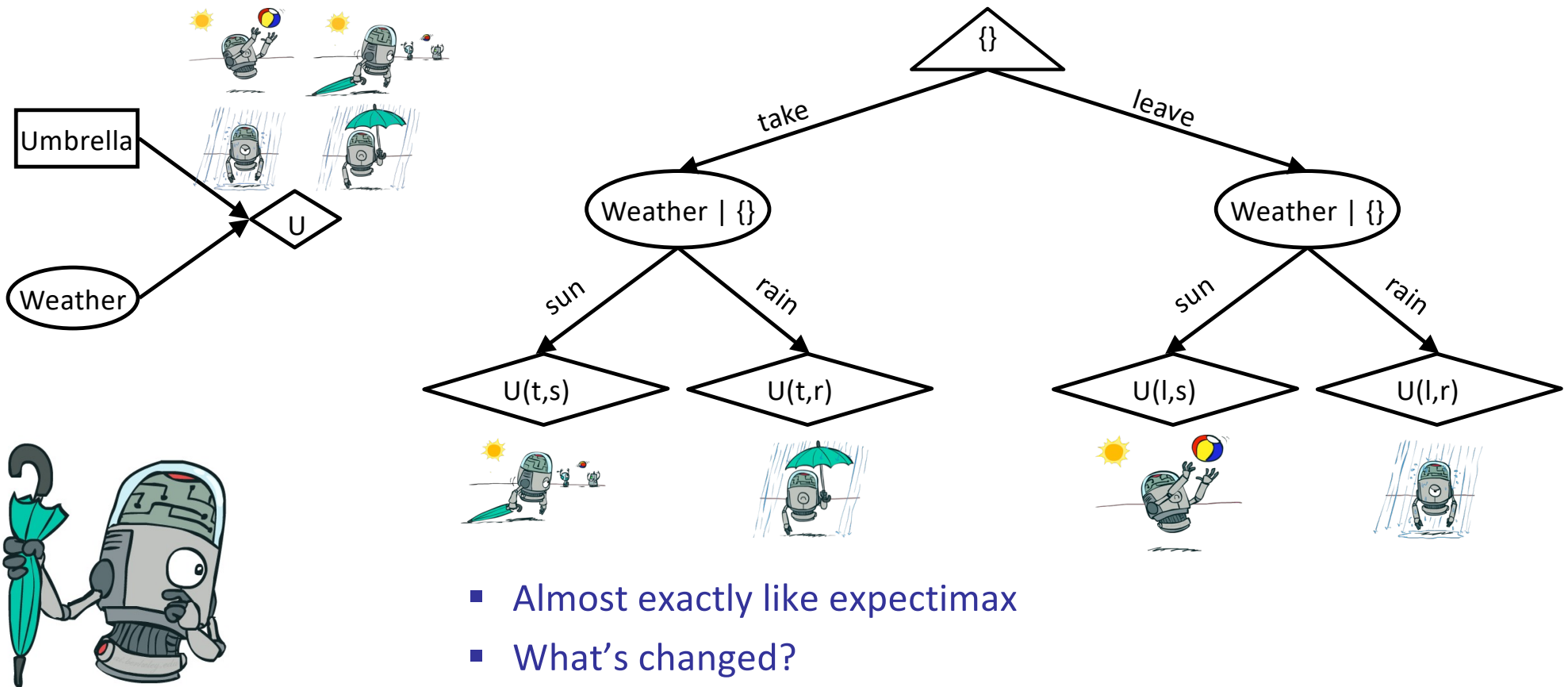
$$\begin{aligned} \text{EU}(\text{take}) &= \sum_w P(w)U(\text{take}, w) \\ &= 0.7 \cdot 20 + 0.3 \cdot 70 = 35 \end{aligned}$$

Optimal decision = leave

$$\text{MEU}(\emptyset) = \max_a \text{EU}(a) = 70$$

- **EU(action) = Expected Utility of taking action**
leave
 - In the parentheses, we write an action
 - Calculating EU requires taking an expectation over chance node outcomes
- **MEU($\emptyset$) = Maximum Expected Utility, given no information**
 - In the parentheses, we write the evidence (which nodes we know)
 - Calculating MEU requires taking a maximum over several expectations (one EU per action)

Decisions as Outcome Trees



Example: Decision Networks

Umbrella = leave

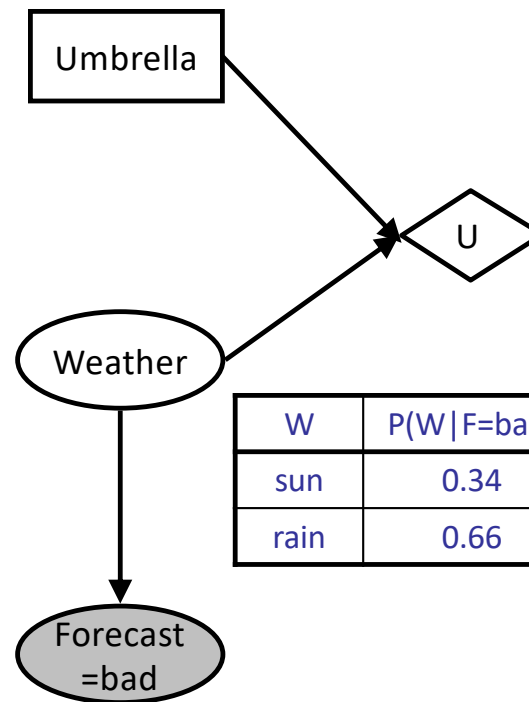
$$\begin{aligned} EU(\text{leave}|\text{bad}) &= \sum_w P(w|\text{bad})U(\text{leave}, w) \\ &= 0.34 \cdot 100 + 0.66 \cdot 0 = 34 \end{aligned}$$

Umbrella = take

$$\begin{aligned} EU(\text{take}|\text{bad}) &= \sum_w P(w|\text{bad})U(\text{take}, w) \\ &= 0.34 \cdot 20 + 0.66 \cdot 70 = 53 \end{aligned}$$

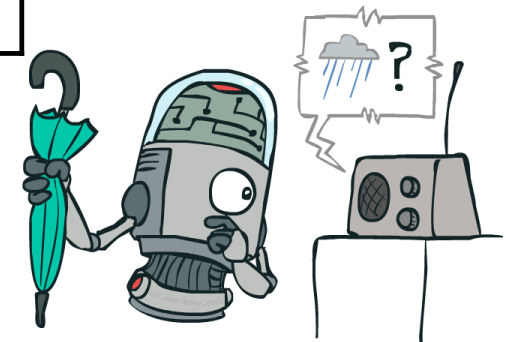
Optimal decision = take

$$MEU(F = \text{bad}) = \max_a EU(a|\text{bad}) = 53$$



A	W	U(A,W)
leave	sun	100
leave	rain	0
take	sun	20
take	rain	70

W	P(W F=bad)
sun	0.34
rain	0.66



Decision Networks: Notation

Umbrella = leave

$$\begin{aligned}\text{EU}(\text{leave}|\text{bad}) &= \sum_w P(w|\text{bad})U(\text{leave}, w) \\ &= 0.34 \cdot 100 + 0.66 \cdot 0 = 34\end{aligned}$$

Umbrella = take

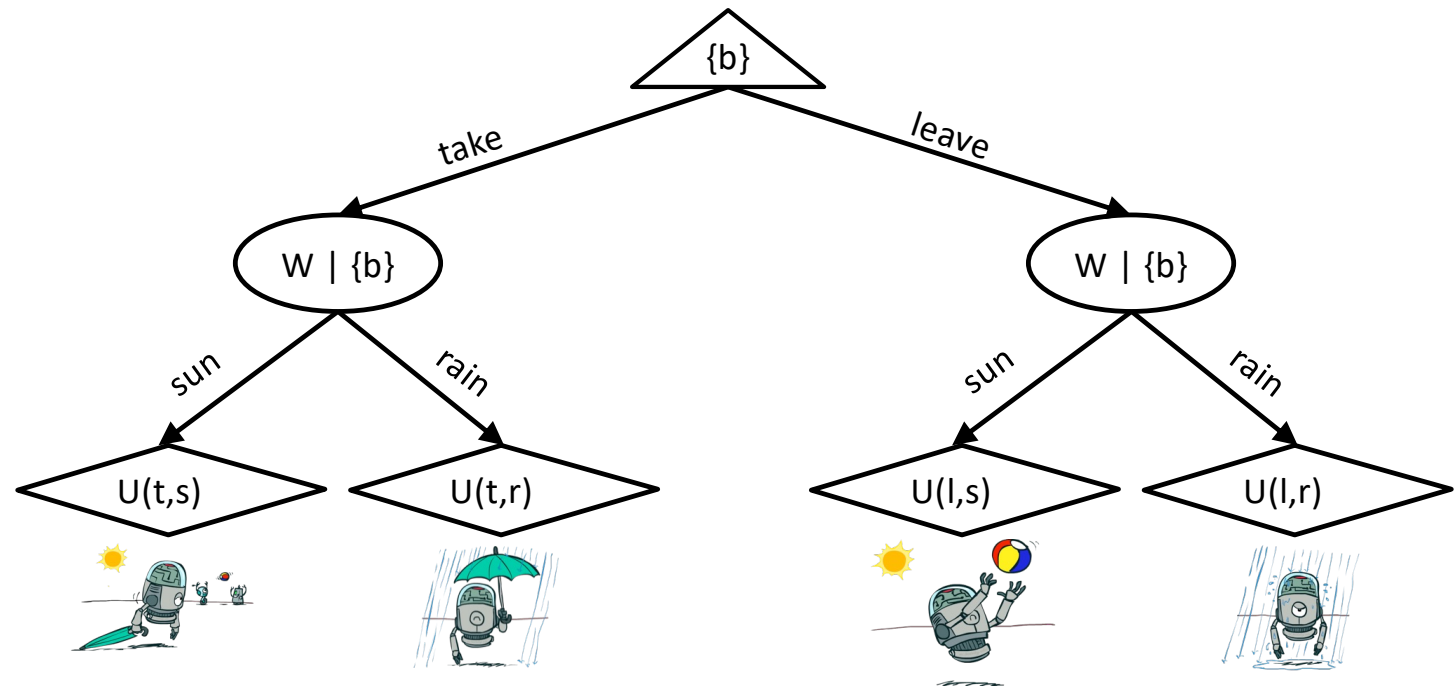
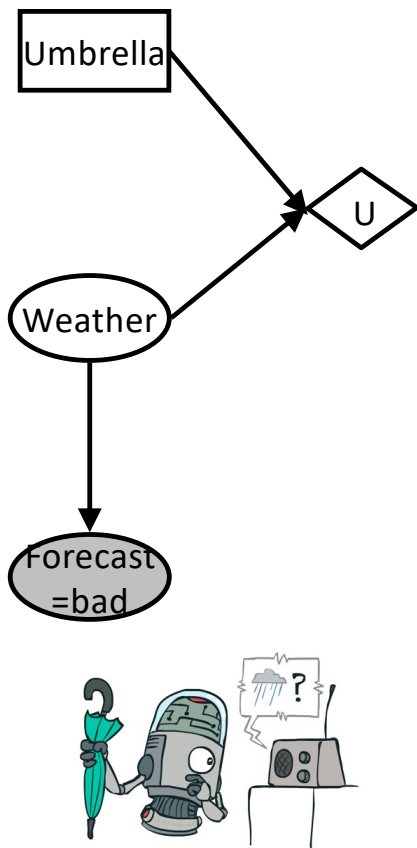
$$\begin{aligned}\text{EU}(\text{take}|\text{bad}) &= \sum_w P(w|\text{bad})U(\text{take}, w) \\ &= 0.34 \cdot 20 + 0.66 \cdot 70 = 53\end{aligned}$$

Optimal decision = take

$$\text{MEU}(F = \text{bad}) = \max_a \text{EU}(a|\text{bad}) = 53$$

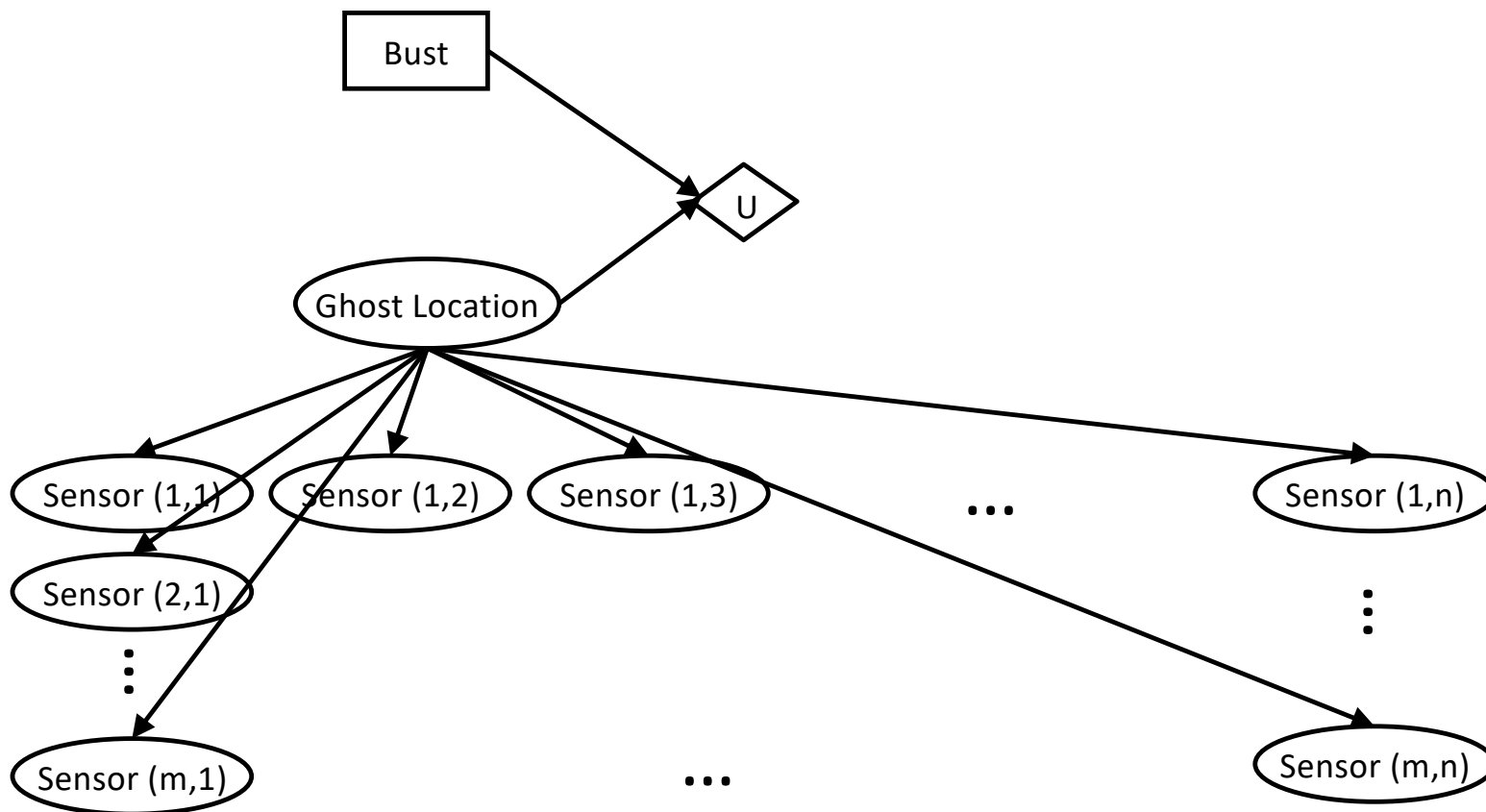
- $\text{EU}(\text{leave}|\text{bad})$ = Expected Utility of choosing leave, given you know the forecast is bad
 - Left side of conditioning bar: Action being taken
 - Right side of conditioning bar: The random variable(s) we know the value of (evidence)
- $\text{MEU}(F=\text{bad})$ = Maximum Expected Utility, given you know the forecast is bad
 - In the parentheses, we write the evidence (which nodes we know)

Decisions as Outcome Trees



Ghostbusters Decision Network

Demo: Ghostbusters with probability



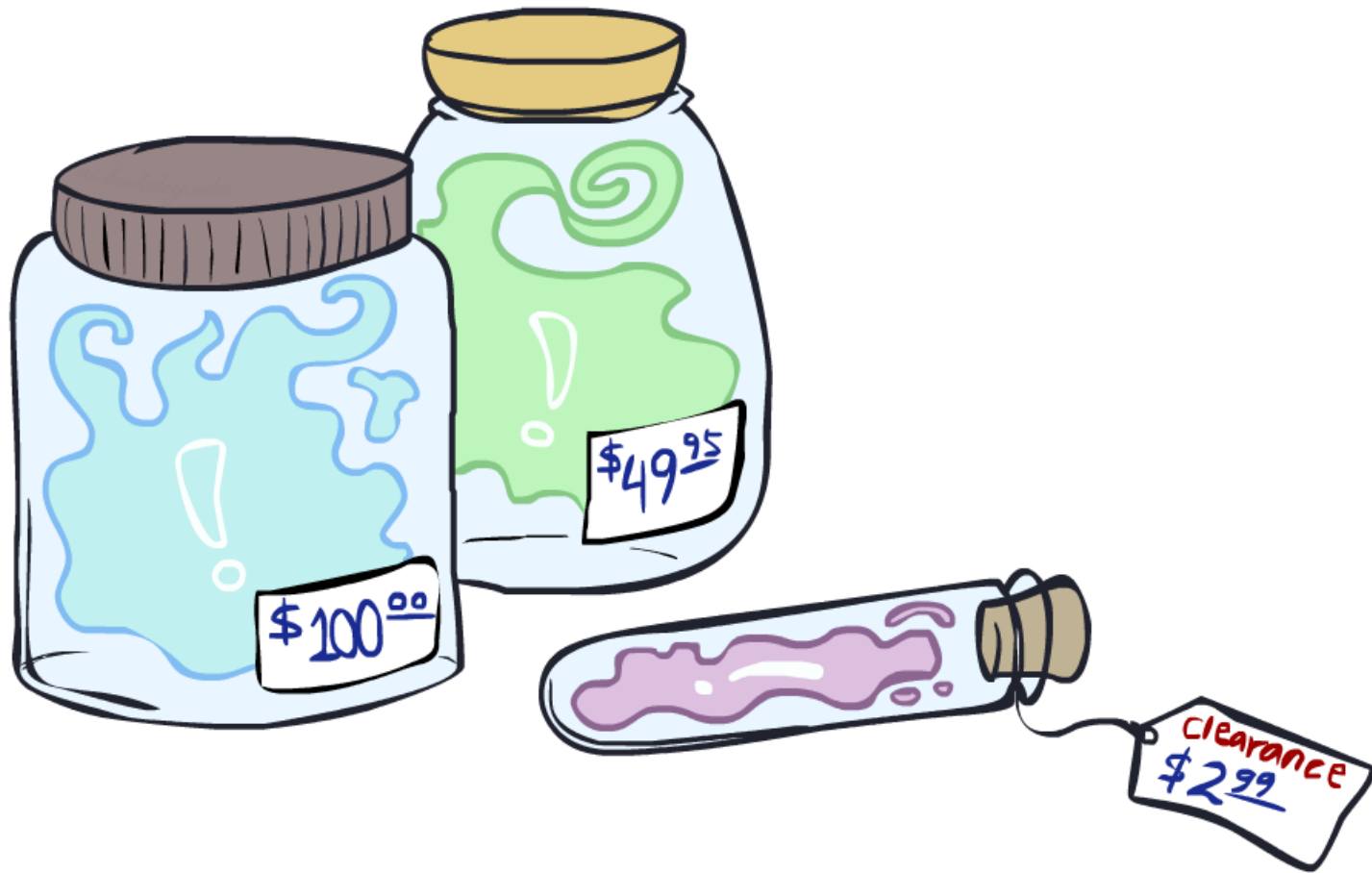
Video of Demo Ghostbusters with Probability

■ Game:

- Costs 1 to make a measurement
- Measurement gives noisy estimate of distance to ghost
- When we blast, game is over
- If we blast the ghost, we get utility of 250

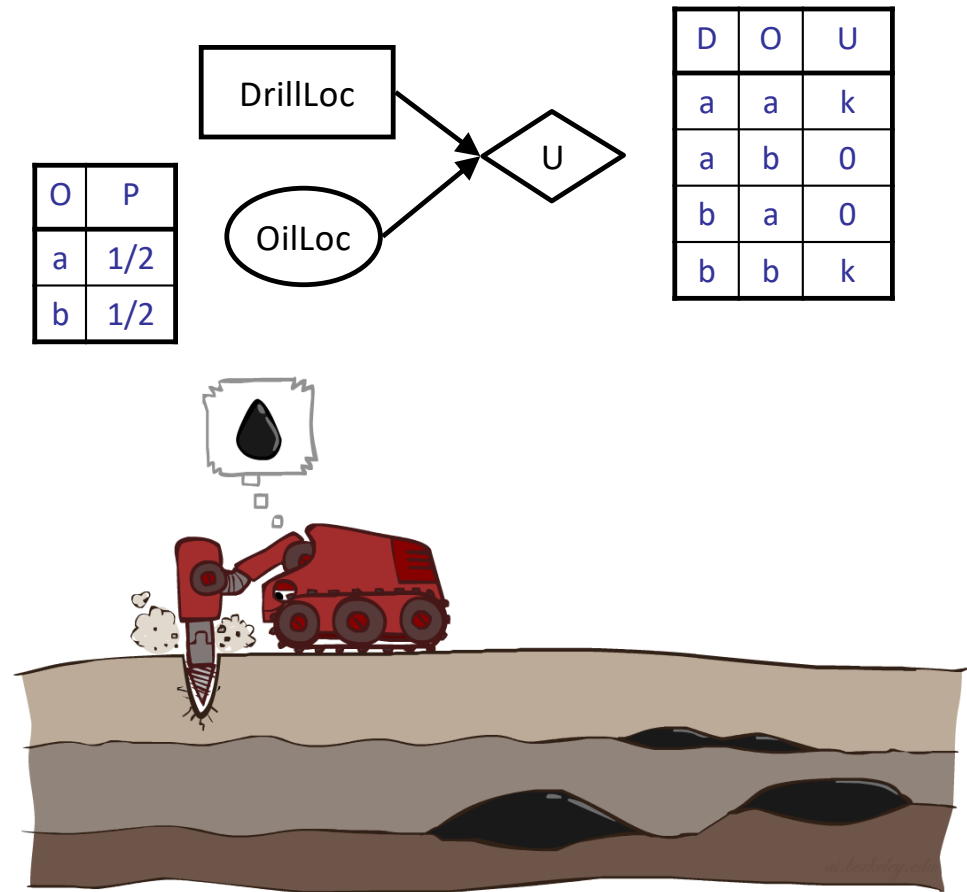


Value of Information



Value of Information

- Idea: compute value of acquiring evidence
 - Can be done directly from decision network
- Example: buying oil drilling rights
 - Two blocks A and B, exactly one has oil, worth k
 - You can drill in one location
 - Prior probabilities 0.5 each, & mutually exclusive
 - Drilling in either A or B has $EU = k/2$, $MEU = k/2$
- Question: what's the value of information of O?
 - Value of knowing which of A or B has oil
 - Value is expected gain in MEU from new info
 - Survey may say "oil in a" or "oil in b"
 - If we know OilLoc, MEU is k (either way)
 - Gain in MEU from knowing OilLoc?
 - $VPI(OilLoc) = k - k/2 = k/2$
 - Fair price of information: $k/2$



Value of Information Example: Weather

MEU with no evidence

$$\text{MEU}(\emptyset) = \max_a \text{EU}(a) = 70$$

MEU if forecast is bad

$$\text{MEU}(F = \text{bad}) = \max_a \text{EU}(a|\text{bad}) = 53$$

argmax_a is "take umbrella"

MEU if forecast is good

$$\text{MEU}(F = \text{good}) = \max_a \text{EU}(a|\text{good}) = 95$$

argmax_a is "leave umbrella"

Forecast distribution

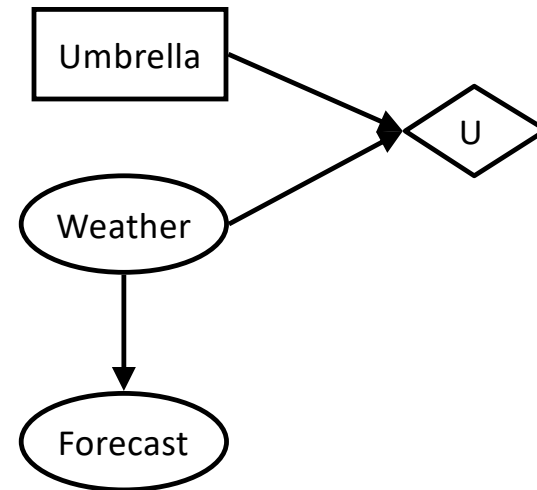
F	P(F)
good	0.59
bad	0.41



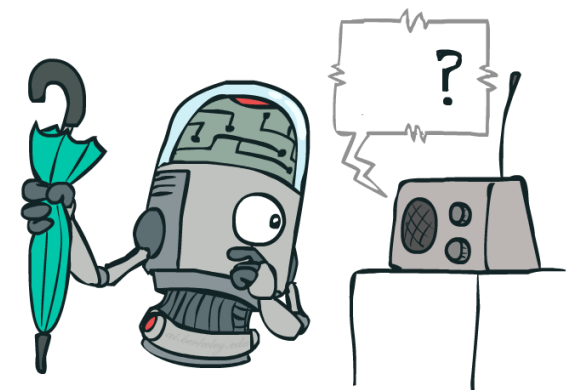
$$\text{MEU}(F) = 0.59 \cdot (95) + 0.41 \cdot (53) = 77.8$$

$$77.8 - 70 = 7.8$$

$$\text{VPI}(E'|e) = \left(\sum_{e'} P(e'|e) \text{MEU}(e, e') \right) - \text{MEU}(e)$$



A	W	U
leave	sun	100
leave	rain	0
take	sun	20
take	rain	70



“Value of Perfect Information”

- Assume we have evidence $E=e$. Value if we act now:

$$MEU(e) = \max_a \sum_s P(s|e) U(s, a)$$

- Assume we see that $E' = e'$. Value if we act then:

$$MEU(e, e') = \max_a \sum_s P(s|e, e') U(s, a)$$

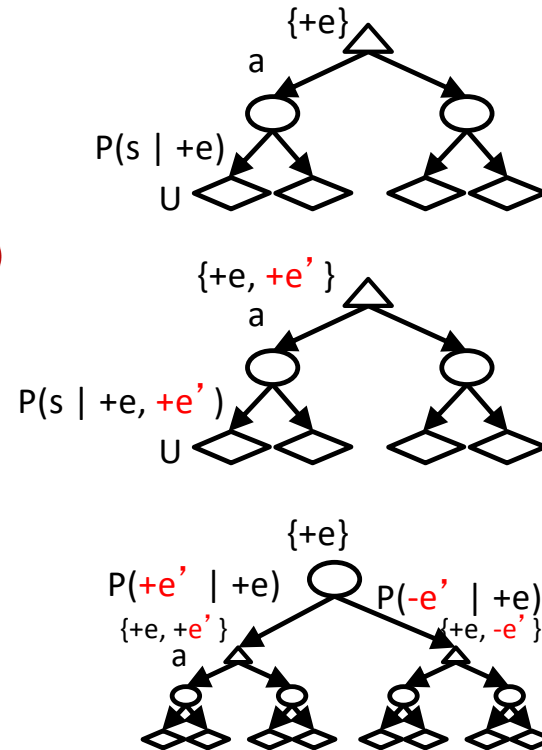
- BUT E' is a random variable whose value is unknown, so we don't know what e' will be

- Expected value if E' is revealed and then we act:

$$MEU(e, E') = \sum_{e'} P(e'|e) MEU(e, e')$$

- Value of information: how much MEU goes up by revealing E' first then acting, over acting now:

$$VPI(E'|e) = MEU(e, E') - MEU(e)$$



VPI: Notation

- $MEU(e)$ = Maximum Expected Utility, given evidence $E=e$
 - In the parentheses, we write the evidence (which nodes we know)
 - Calculating MEU requires taking a maximum over several expectations (one EU per action)
- $VPI(E'|e)$ = Expected gain in utility for knowing the value of E' , given that I know the value of e so far
 - Left side of conditioning bar: The random variable(s) we want to know the value of revealing
 - Right side of conditioning bar: The random variable(s) we already know the value of
 - Calculating VPI requires taking an expectation over several MEUs (one MEU per possible outcome of E' , because we don't know the value of E')

$$MEU(e) = \max_a \sum_s P(s|e) U(s, a)$$
$$MEU(e, e') = \max_a \sum_s P(s|e, e') U(s, a)$$
$$VPI(E'|e) = \left(\sum_{e'} P(e'|e) MEU(e, e') \right) - MEU(e)$$

VPI: Computation Workflow

$$\text{MEU}(e) = \max_a \text{EU}(a|e)$$

$$\text{MEU}(e, e') = \max_a \text{EU}(a|e, e') \quad (\text{calculate this for all values } e' \text{ that } E' \text{ could take})$$

$$\text{MEU}(e, E') = \sum_{e'} P(e'|e) \text{MEU}(e, e')$$

$$\text{MEU}(e, E') - \text{MEU}(e) = \text{VPI}(E'|e)$$

Video of Demo Ghostbusters with VPI

■ Game:

- Costs 1 to make a measurement
- Measurement gives noisy estimate of distance to ghost
- When we blast, game is over
- If we blast the ghost, we get utility of 250



VPI Properties

- Nonnegative

$$\forall E', e : \text{VPI}(E'|e) \geq 0$$

(Positive if different observed values of e' lead to different optimal decisions)



- Subadditive

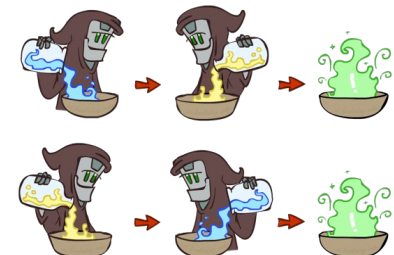
$$\text{VPI}(E_j, E_k|e) \leq \text{VPI}(E_j|e) + \text{VPI}(E_k|e)$$

(think of observing the same E_j twice)



- Order-independent

$$\begin{aligned} \text{VPI}(E_j, E_k|e) &= \text{VPI}(E_j|e) + \text{VPI}(E_k|e, E_j) \\ &= \text{VPI}(E_k|e) + \text{VPI}(E_j|e, E_k) \end{aligned}$$

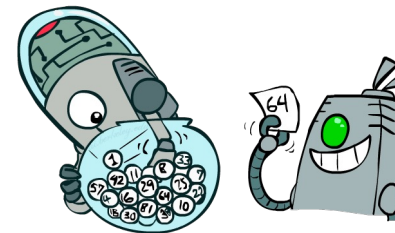
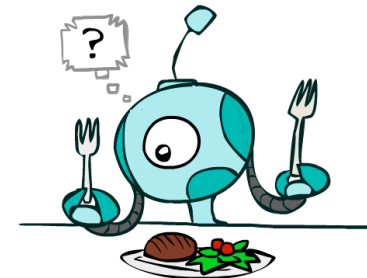
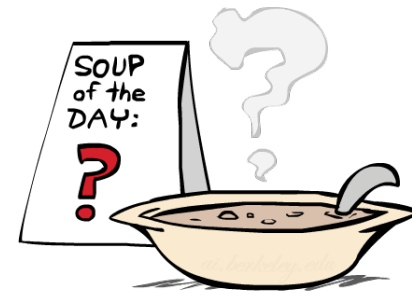


Value of information contd.

- General idea: value of information = *expected improvement in decision quality* from observing value of a variable
 - E.g., oil company deciding on seismic exploration and test drilling
 - E.g., doctor deciding whether to order a blood test
 - E.g., person deciding on whether to look before crossing the road
- *Decision network contains everything needed to compute it!*
- $VPI(E_i | e) = \left[\sum_{e_i} P(e_i | e) \max_a EU(a | e_i, e) \right] - \max_a EU(a | e)$

Quick VPI Questions

- The soup of the day is either clam chowder or split pea, but you wouldn't order either one. What's the value of knowing which it is?
- There are two kinds of plastic forks at a picnic. One kind is slightly sturdier. What's the value of knowing which?
- You're playing the lottery. The prize will be \$0 or \$100. You can play any number between 1 and 100 (chance of winning is 1%). What is the value of knowing the winning number?



Value of Imperfect Information?

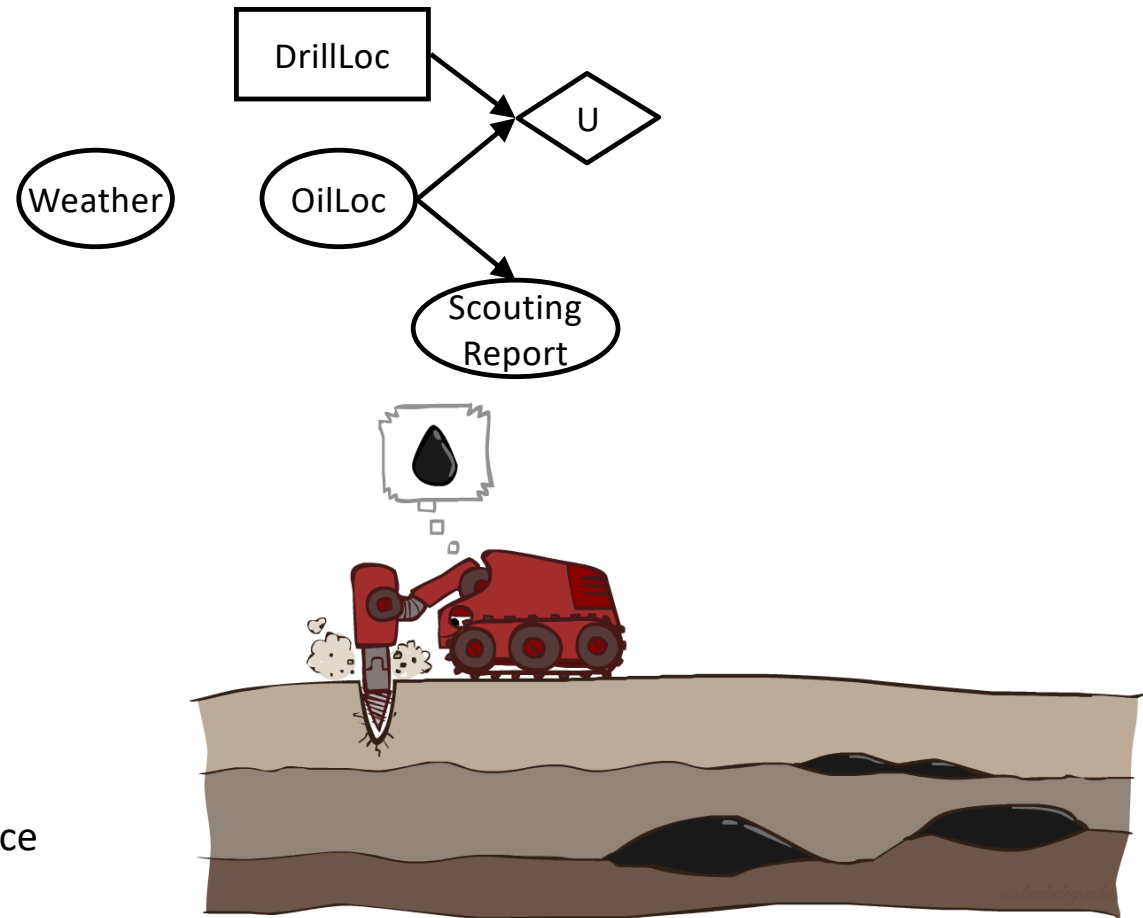


- No such thing (as we formulate it)
- Information corresponds to the observation of a node in the decision network
- If data is “noisy” that just means we don’t observe the original variable, but another variable which is a noisy version of the original one

VPI Question

- $VPI(\text{ScoutingReport})$?
- $VPI(\text{OilLoc})$?
- $VPI(\text{Weather})$?
- $VPI(\text{OilLoc} \mid \text{ScoutingReport})$ vs $VPI(\text{ScoutingReport} \mid \text{OilLoc})$?
- Generally:

$$VPI(Z \mid \text{CurrentEvidence}) = 0$$
 if $\text{Parents}(U) \perp\!\!\!\perp Z \mid \text{CurrentEvidence}$



Bonus slide (if time)

Post-decision Disappointment: the Optimizer's Curse

- Usually we don't have direct access to exact utilities, only *estimates*
 - E.g., you could make one of k investments
 - An unbiased expert assesses their expected net profit $V_1, \dots, V_k$
 - You choose the best one V^*
 - With high probability, *its actual value is considerably less* than V^*
- This is a serious problem in many areas:
 - Future performance of mutual funds
 - Efficacy of drugs measured by trials
 - Statistical significance in scientific papers
 - Winning an auction

Suppose true net profit is 0
and estimate $\sim N(0,1)$;
Max of k estimates:

